

Docker Networking Lab

Custom Bridge Networks — Container DNS — Network Isolation

Scenario 1: Containers created with network

Scenario 2: Running containers connected to network

Bonus: `--network none` isolation

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● syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$ docker network ls

NETWORK ID	NAME	DRIVER	SCOPE
fba6f2f6e2c7	bridge	bridge	local
d843fe92e413	host	host	local
d85eb93ae557	none	null	local

○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$

0 0 5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 5

- syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$ docker network create mynetwork
dc6a8910d071774ef95871d26cfb4ee346f5e247b23f22c3f55015f97c589766
- syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$ docker network ls

NETWORK ID	NAME	DRIVER	SCOPE
fba6f2f6e2c7	bridge	bridge	local
4c4560fc0c87	devops-net	bridge	local
d843fe92e413	host	host	local
dc6a8910d071	mynetwork	bridge	local
d85eb93ae557	none	null	local
- syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$

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d85eb93ae557  none      null      local
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker run -d --name container1 nginx
docker run -d --name container2 nginx
98df6006c585f2c4e3c15ce7dfbdae097fe410ad8f98550c3dccb4050ea3462a
44e532f393c1f81251925248c1f4f06ec8fa034697ba537d2ba850ac47cdc8c6
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
44e532f393c1   nginx         "/docker-entrypoint...."  5 seconds ago  Up 5 seconds  80/tcp                             container2
98df6006c585   nginx         "/docker-entrypoint...."  6 seconds ago  Up 5 seconds  80/tcp                             container1
ea635a63d49e   nginx         "/docker-entrypoint...."  2 hours ago    Up 2 hours    80/tcp                             web2
e59cd82131f5   nginx         "/docker-entrypoint...."  2 hours ago    Up 2 hours    80/tcp                             web1
991ebe3b1621   nginx         "/docker-entrypoint...."  2 hours ago    Up 2 hours                                isolated-container
816ebd232a43   postgres     "docker-entrypoint.s..."  2 hours ago    Up 2 hours    0.0.0.0:5444->5432/tcp, [::]:5444->5432/tcp  pg-lab-2
4a57c1b56d6d   postgres     "docker-entrypoint.s..."  25 hours ago   Up 25 hours    0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp  mydb
a1cb16ad479c   react-multistage:v1 "/docker-entrypoint...."  25 hours ago   Up 25 hours    0.0.0.0:8083->80/tcp, [::]:8083->80/tcp      My-New-Cont
ac4e7eb8825b   react-multistage:v1 "/docker-entrypoint...."  2 days ago     Up 2 days      0.0.0.0:8082->80/tcp, [::]:8082->80/tcp      volume-demo3
b14a9b6e900f   react-multistage:v1 "/docker-entrypoint...."  2 days ago     Up 2 days      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp      react-app
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes$

```

```
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
^C
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker network connect mynetwork container1
docker network connect mynetwork container2
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker network inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "dc6a8910d071774ef95871d26cfb4ee346f5e247b23f22c3f55015f97c589766",
    "Created": "2026-06-17T00:16:31.135411799-04:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "44e532f393c1f81251925248c1f4f06ec8fa034697ba537d2ba850ac47cdc8c6": {
        "Name": "container2",
        "EndpointID": "5887286d5c55299ac21142418217d8f95b993da8e55c0ea3262573f058e14c5c",
        "MacAddress": "c6:01:93:34:6d:1f",
        "IPv4Address": "172.19.0.3/16",
        "IPv6Address": ""
      }
    }
  },
]
```

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        {
            "Subnet": "172.19.0.0/16",
            "Gateway": "172.19.0.1"
        }
    ],
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
        "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
        "44e532f393c1f81251925248c1f4f06ec8fa034697ba537d2ba850ac47cdc8c6": {
            "Name": "container2",
            "EndpointID": "5887286d5c55299ac21142418217d8f95b993da8e55c0ea3262573f058e14c5c",
            "MacAddress": "c6:01:93:34:6d:1f",
            "IPv4Address": "172.19.0.3/16",
            "IPv6Address": ""
        },
        "98df6006c585f2c4e3c15ce7dfbdae097fe410ad8f98550c3dccf4050ea3462a": {
            "Name": "container1",
            "EndpointID": "34573b745bd218f1477afc0b1103dd96d3a68f5de787b0f0b8b18a10a284e0ed",
            "MacAddress": "be:93:76:d7:31:2a",
            "IPv4Address": "172.19.0.2/16",
            "IPv6Address": ""
        }
    },
    "Status": {
        "IPAM": {
            "Subnets": {
                "172.19.0.0/16": {
                    "IPsInUse": 5,
                    "DynamicIPsAvailable": 65531
                }
            }
        }
    }
}
]

```

o syedaftab04@SYEDBASIT26:~/docker-labs/volumes\$


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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 5
syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker inspect devops-net
[
  {
    "Created": "2026-06-16T22:42:46.667671803-04:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": [
      {
        "e59cd82131f5ae6ab021fed3bcc9118201729bf3949b50bfd781d77271e7b15b": {
          "Name": "web1",
          "EndpointID": "fcdcf2862cad006d8679af81269d9018d41eed7e5b68dd86fcd7edfe9c99641b3",
          "MacAddress": "7a:ac:ba:ac:4c:74",
          "IPv4Address": "172.18.0.2/16",
          "IPv6Address": ""
        },
        {
          "ea635a63d49ea426fd34858b96023f0569318421d35d039fc358790948193aff": {
            "Name": "web2",
            "EndpointID": "9e23a86327798faa4651b6077d3fc462439c173236b00e5c6416b18dc2e48be1",
            "MacAddress": "62:7f:60:5d:bc:6b",
            "IPv4Address": "172.18.0.3/16",
            "IPv6Address": ""
          }
        }
      ]
    },
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "172.18.0.0/16": {

```

```
}  
}  
]  
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker exec -it container1 bash  
root@98df6006c585:/# ping container2  
PING container2 (172.19.0.3) 56(84) bytes of data.  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=1 ttl=64 time=2.74 ms  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=2 ttl=64 time=0.064 ms  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=3 ttl=64 time=0.072 ms  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=4 ttl=64 time=0.107 ms  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=5 ttl=64 time=0.107 ms  
64 bytes from container2.mynetwork (172.19.0.3): icmp_seq=6 ttl=64 time=0.162 ms  
^C  
--- container2 ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5074ms  
rtt min/avg/max/mdev = 0.064/0.541/2.738/0.982 ms  
root@98df6006c585:/#
```

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```
}  
}  
}  
}  
]  
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker exec -it web2 sh  
# ls  
bin boot dev docker-entrypoint.d docker-entrypoint.sh etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var  
# echo "Hello from Web2" > /usr/share/nginx/html/index.html  
exit# cat /usr/share/nginx/html/index.html  
sh: 3: exitcat: not found  
# cat /usr/share/nginx/html/index.html  
Hello from Web2  
# exit  
• syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ /usr/share/nginx/html/index.html  
bash: /usr/share/nginx/html/index.html: No such file or directory  
❖ syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker exec -it web1 sh  
# apt update  
Hit:1 http://deb.debian.org/debian trixie InRelease  
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]  
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease  
Fetched 47.3 kB in 0s (357 kB/s)  
All packages are up to date.  
# apt update  
apt install curl -yError: Command line option 'a' [from -yapt] is not understood in combination with the other options.
```

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```

d85eb93ae557 none null local
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker network create devops-net
4c4560fc0c878377b416bb4fa9dd8fa2f379767a37378370695f8701b12c73ca
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker network ls
NETWORK ID      NAME      DRIVER  SCOPE
fba6f2f6e2c7    bridge    bridge   local
4c4560fc0c87    devops-net bridge   local
d843fe92e413    host      host     local
d85eb93ae557    none      null     local
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker run -d \
--name web1 \
--network devops-net \
nginx
e59cd82131f5ae6ab021fed3bcc9118201729bf3949b50bfd781d77271e7b15b
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker run -d \
--name web2 \
--network devops-net \
nginx
ea635a63d49ea426fd34858b96023f0569318421d35d039fc358790948193aff
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker ps

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CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ea635a63d49e	nginx	"/docker-entrypoint..."	5 seconds ago	Up 5 seconds	80/tcp	web2
e59cd82131f5	nginx	"/docker-entrypoint..."	19 seconds ago	Up 18 seconds	80/tcp	web1
991ebe3b1621	nginx	"/docker-entrypoint..."	15 minutes ago	Up 15 minutes		isolated-container
816ebd232a43	postgres	"docker-entrypoint.s..."	52 minutes ago	Up 52 minutes	0.0.0.0:5444->5432/tcp, [::]:5444->5432/tcp	pg-lab-2
4a57c1b56d6d	postgres	"docker-entrypoint.s..."	23 hours ago	Up 23 hours	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp	mydb
a1cb16ad479c	react-multistage:v1	"/docker-entrypoint..."	23 hours ago	Up 23 hours	0.0.0.0:8083->80/tcp, [::]:8083->80/tcp	My-New-Cont
ac4e7eb8825b	react-multistage:v1	"/docker-entrypoint..."	2 days ago	Up 2 days	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp	volume-demo3
b14a9b6e900f	react-multistage:v1	"/docker-entrypoint..."	2 days ago	Up 2 days	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	react-app

```

● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$

```

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o syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker exec -it web1 sh
# apt update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [216 kB]
Fetched 10.1 MB in 1s (9908 kB/s)
All packages are up to date.
# apt install curl -y
Error: Command line option 'a' [from -yapt] is not understood in combination with the other options.
# apt install curl -y
curl is already the newest version (8.14.1-2+deb13u3).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
# curl http://web2
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
#
```



```
Run 'docker run --help' for more information
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker run -d \
  --name isolated-container \
  --network none \
  nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d4dcde3aeed: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
57b3fbf43092: Pull complete
6376488be516: Pull complete
c5a7565de4cf: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
991ebe3b1621b70e13273057aa5a22a8136d4c4aa906bbd77e487a5bb876b0cd
● syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker inspect isolated-container | grep NetworkMode
    "NetworkMode": "none",
○ syedaftab04@SYEDBASIT26:~/docker-labs/volumes$
```

0 0 5



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syedaftab04@SYEDBASIT26:~/docker-labs/volumes$ docker inspect pg-lab-2
  "StopSignal": "SIGINT"
},
  "NetworkSettings": {
    "SandboxID": "ffc1abaaa15d15344cb4fdb904a032b15f042434823a75465a0c39db77a9ed27",
    "SandboxKey": "/var/run/docker/netns/ffc1abaaa15d",
    "Ports": {
      "5432/tcp": [
        {
          "HostIp": "0.0.0.0",
          "HostPort": "5444"
        },
        {
          "HostIp": "::",
          "HostPort": "5444"
        }
      ]
    },
    "Networks": {
      "bridge": {
        "IPAMConfig": null,
        "Links": null,
        "Aliases": null,
        "DriverOpts": null,
        "GwPriority": 0,
        "NetworkID": "fba6f2f6e2c7c7a8f43cb846f447b50a5b9e115ad6d92cca2929cbd239e5bfff0",
        "EndpointID": "be91c410eb4d392097749779abaa6a0c60b98a76b413b33ca0f1e3265dca489d",
        "Gateway": "172.17.0.1",
        "IPAddress": "172.17.0.6",
        "MacAddress": "f6:d9:b8:50:e7:d2",
        "IPPrefixLen": 16,
        "IPv6Gateway": "",
        "GlobalIPv6Address": "",
        "GlobalIPv6PrefixLen": 0,
        "DNSNames": null
      }
    }
  },
},

```